

Regres_logis040726

July 4, 2026

```
[1]: # Librerias

import pandas as pd
import numpy as np
from sklearn import linear_model
from sklearn import model_selection
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
```

```
[2]: # Cargar datos
data = pd.read_csv('usuarios_win_mac_lin.csv')
```

```
[3]: # Ver los primeros renglones con head
data.head()
```

```
[3]:
```

	duracion	paginas	acciones	valor	clase
0	7.0	2	4	8	2
1	21.0	2	6	6	2
2	57.0	2	4	4	2
3	101.0	3	6	12	2
4	109.0	2	6	12	2

```
[4]: # Revisar forma
data.shape
```

```
[4]: (170, 5)
```

```
[5]: # Estadística descriptiva
data.describe()
```

```
[5]:
```

	duracion	paginas	acciones	valor	clase
count	170.000000	170.000000	170.000000	170.000000	170.000000
mean	111.075729	2.041176	8.723529	32.676471	0.752941
std	202.453200	1.500911	9.136054	44.751993	0.841327
min	1.000000	1.000000	1.000000	1.000000	0.000000
25%	11.000000	1.000000	3.000000	8.000000	0.000000
50%	13.000000	2.000000	6.000000	20.000000	0.000000

75%	108.000000	2.000000	10.000000	36.000000	2.000000
max	898.000000	9.000000	63.000000	378.000000	2.000000

```
[6]: # Ver cuantos renglones tengo de cada clase (sistemas operativos)
data.groupby('clase').size()
```

```
[6]: clase
0      86
1      40
2      44
dtype: int64
```

```
[7]: # Preparación de mis datos, matriz de entrenamiento "X" que contiene las
      ↪ variables independientes y un vector de valores (labels) "y" que contiene la
      ↪ variable dependiente
X = np.array(data.drop(['clase'], axis = 1))
```

```
[8]: y = np.array(data['clase'])
```

```
[9]: X.shape
```

```
[9]: (170, 4)
```

```
[10]: # Crear modelo predictivo
model = linear_model.LogisticRegression()
```

```
[11]: # Entrenar modelo
model.fit(X,y)
```

```
/Users/haydeml/Library/Python/3.9/lib/python/site-
packages/sklearn/linear_model/_logistic.py:469: ConvergenceWarning: lbfgs failed
to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.
```

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
```

```
[11]: LogisticRegression()
```

```
[12]: # Crear predicciones
predictions = model.predict(X)
```

```
[13]: predictions
```

```
[13]: array([2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 0, 0, 0, 0, 1,
            0, 1, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 2, 2, 2, 2, 2, 0, 0, 0, 1,
            0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0,
            0, 1, 0, 0, 0, 0, 2, 1, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 2, 0, 2, 0,
            0, 0, 0, 0, 0, 0, 2, 0, 2, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1,
            2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 0, 0,
            0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 2, 0, 0,
            0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0])
```

```
[14]: y
```

```
[14]: array([2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 1, 1, 1, 1, 1,
            1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 1, 1, 1, 1,
            1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
            0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
            0, 0, 0, 0, 0, 0, 2, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
            2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 0, 0,
            0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
            0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0])
```

```
[15]: model.score(X,y)
```

```
[15]: 0.7823529411764706
```

```
[16]: # BUENAS PRACTICAS
# Separar el dataset en un conjunto de entrenamiento y un conjunto de
    ↪ validación o prueba
validation_size = 0.20
seed = 7
X_train, X_validation, y_train, y_validation = model_selection.
    ↪ train_test_split(X,y, test_size = validation_size, random_state = seed)
```

```
[22]: # Entrenar de nuevo el modelo
name = 'Logistic Regression'
kfold = model_selection.KFold(n_splits=10)
cv_results = model_selection.cross_val_score(model, X_train, y_train, cv=kfold,
    ↪ scoring= 'accuracy')
msg = name, ": ", cv_results.mean(), "(", cv_results.std()
print(msg)
```

```
('Logistic Regression', ': ', 0.7285714285714284, '(', 0.09418550477897196)
```

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```
n_iter_i = _check_optimize_result(
```

```
[24]: # Predicción del conjunto de validación
      predictions = model.predict(X_validation)
```

```
[25]: # Evaluar el modelo con los datos que no vio en el entrenamiento
      accuracy_score(y_validation, predictions)
```

```
[25]: 0.8529411764705882
```

```
[26]: # Matriz de confusión
      confusion_matrix(y_validation, predictions)
```

```
[26]: array([[16,  0,  2],
           [ 3,  3,  0],
           [ 0,  0, 10]])
```

```
[27]: print(classification_report(y_validation, predictions))
```

	precision	recall	f1-score	support
0	0.84	0.89	0.86	18
1	1.00	0.50	0.67	6
2	0.83	1.00	0.91	10
accuracy			0.85	34
macro avg	0.89	0.80	0.81	34
weighted avg	0.87	0.85	0.84	34

```
[28]: # Predicción de un evento simulado de alguien que entró al sitio web
      X_new = pd.DataFrame({'duracion':[10], 'paginas':[2], 'acciones':[5], 'valor':
      ↪ [9]})
```

```
[29]: model.predict(X_new)
```

```
/Users/haydeml/Library/Python/3.9/lib/python/site-packages/sklearn/base.py:486:
UserWarning: X has feature names, but LogisticRegression was fitted without
feature names
  warnings.warn(
```

```
[29]: array([2])
```

[]: